

Assignment 1: azure virtual network setup

Objective: create a virtual network which has multiple subnets:web,App, Db and mgmt and secure the subnets by implementing the network security groups.

create virtual network with 4 subnets

Virtual Network

- Name: azurenetworkproj1
- Address Space: 10.1.0.0/16

Subnets Created

Subnet	Address Range
-----	-----
Web	10.1.0.0/24
App	10.1.2.0/24
DB	10.1.3.0/24
Mgmt	10.1.4.0/24

Now implement Network security groups for subnets

Web NSG:Allow traffic ONLY from Mgmt subnet

Source: 10.1.4.0/24 (Mgmt subnet)

Destination: Web subnet

App NSG:allow traffic Only Web + Mgmt can access

- Source: 10.1.4.0/24 (Mgmt subnet)

-Source: 10.1.1.0/24 (web subnet)

DB NSG:Only App + Mgmt access

-Source: 10.0.2.0/24 (App subnet)

-Source: 10.0.4.0/24 (Mgmt subnet)

Mgmt NSG:traffic from only our own IP or public

- source:any

Now create virtual machine for each subnet by configuring NSG and remaining all default settings

- Web VM → Web subnet-webnsg
- App VM → App subnet-appnsg
- DB VM → DB subnet-dbnsng
- Mgmt VM → Mgmt subnet-mgmtnsg

Architecture Flow

User → Mgmt → Web → App → DB
Access all servers

Security Design

- No public access to Web/App/DB
- Only Mgmt VM exposed to internet

Key Learnings

- Implemented 3-tier architecture
- Used subnet-based network segmentation
- Configured jump server for secure access

Day 1 Presentation - Azure Networking

Architecture

[Internet

↓

[Mgmt (Jump Server)]

↓

[Web Layer]

↓

[App Layer]

↓

[DB Layer]

Implementation:

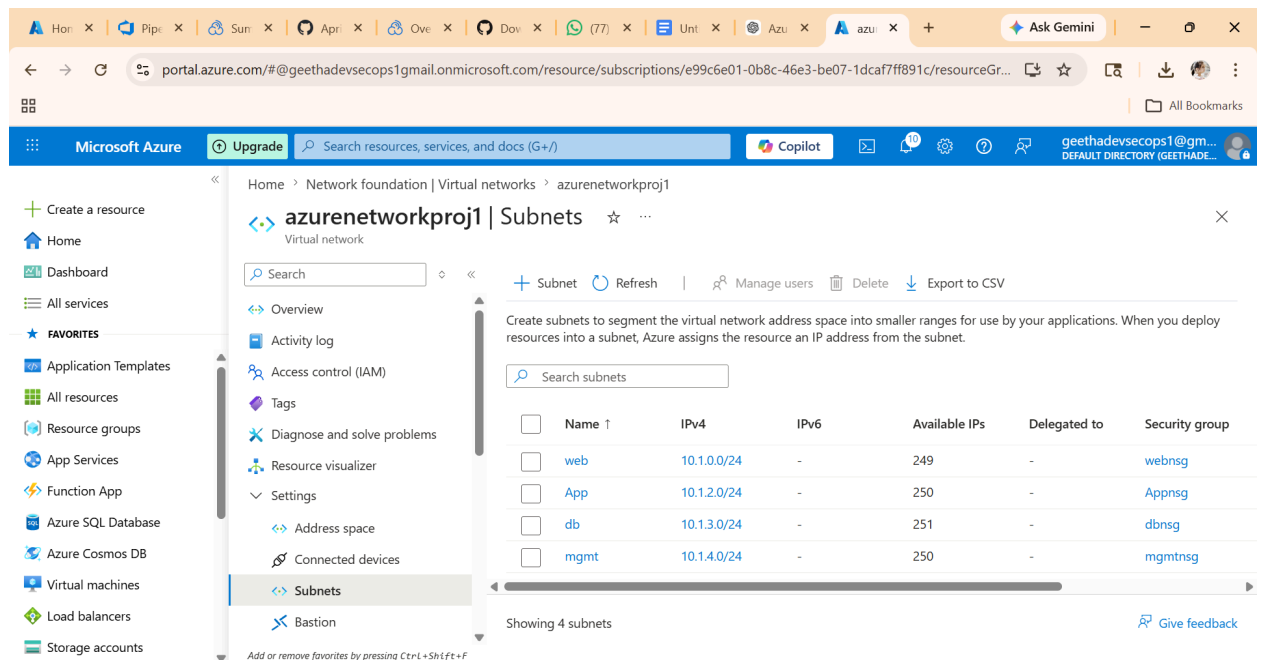
- Created Virtual Network (10.0.0.0/16)
- Created 4 subnets (Web, App, DB, Mgmt)
- Deployed 4 VMs mapped to subnets
- Configured Network Security Groups
- Implemented Jump Server architecture

Security Approach:

- Only Mgmt VM has Public IP
- Internal VMs are private
- NSGs restrict access between layers

Azure portal screenshots:

- VNet with subnets



- NSGs

Microsoft Azure portal screenshot showing the Network foundation | Network security groups page. The page displays a list of network security groups (NSGs) associated with the resource group 'geethadevops' in the 'South India' location. The table shows the following data:

Name	Resource Group	Location	Subscription	Flow log
Appnsg	geethadevops	South India	Azure subscript...	
dbnsg	geethadevops	South India	Azure subscript...	
mgmntnsg	geethadevops	South India	Azure subscript...	
webnsg	geethadevops	South India	Azure subscript...	

The page also includes a sidebar with navigation options, a search bar, and a 'Give feedback' link.

- Virtual machines:

Microsoft Azure portal screenshot showing the Compute infrastructure | Virtual machines page. The page displays a list of virtual machines (VMs) associated with the resource group 'geethadevops' in the 'South India' location. The table shows the following data:

Name	Subscription	Resource Group	Location	Status
appmachine	Azure subscript...	geethadevops	South India	Stopped (deall...
jump-VM	Azure subscript...	geethadevops	South India	Running

The page also includes a sidebar with navigation options, a search bar, and a 'Give feedback' link.

Validation:

Connect to jump server: and then from connect to other servers web server, app server and db server.

```
azureuser@jump-VM: ~
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

System information as of Mon Apr 27 19:11:14 UTC 2026

System load: 0.0      Processes:      126
Usage of /:  7.3% of 28.02GB Users logged in: 0
Memory usage: 38%    IPv4 address for eth0: 10.1.4.4
Swap usage:  0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

3 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Apr 21 20:35:19 2026 from 49.37.145.77
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@jump-VM:~$
```

```
Windows PowerShell

* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

System information as of Tue Apr 21 20:36:50 UTC 2026

System load: 0.08      Processes:      125
Usage of /:  5.7% of 28.02GB Users logged in: 0
Memory usage: 32%    IPv4 address for eth0: 10.1.2.4
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@apppmachine:~$ exit
logout
Connection to 10.1.2.4 closed.
azureuser@jump-VM:~$ exit
logout
Connection to 20.219.127.23 closed.
PS C:\WINDOWS\system32> ssh azureuser@10.1.2.4
ssh: connect to host 10.1.2.4 port 22: Connection timed out
PS C:\WINDOWS\system32>
```

Issues faced:

Since I am using azure free trial account, the maximum virtual machines that can be created for a region is limited to 2 as the maximum quota of vcpus is 4, so I created only 2 virtual machines and validated.